universidade de aveiro

instituto de telecomunicações

Roaming For Applications

M2 - Grupo 14

125046 Alan Marques
125011 Margarida Pinheiro
125274 Rodrigo Sampaio
125624 Miguel Sousa
119625 João Ribeiro

2026

Recordar o Contexto

Necessidade



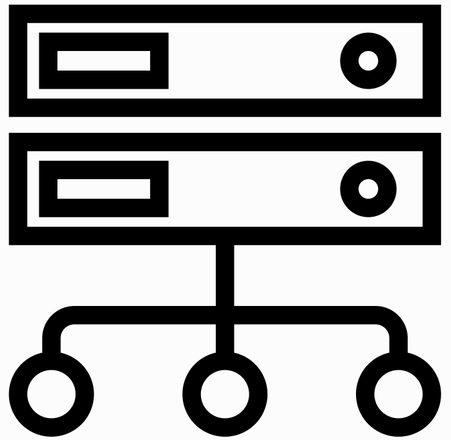
A aplicação precisa de recursos próximos

Obstáculo



O operador pode não os ter

Solução



Um parceiro pode disponibilizá-los

STATE OF THE ART

	LIQ	GSMA	IPCEI-CIS	OpenOP by ETSI	Our Project
Inter-Operator Federation	✗	✓	✓	✓	✓
Latency & Location Aware Decision	✗	✓	✓	✗	✓
Transparent Application Handover	✗	✓	✓	✗	✓
Single Developer Entry Point (PAP)	✓	✗	✓	✓	✓
Executable Prototype	✓	✗	✗	✓	✓

Requirements Gathering



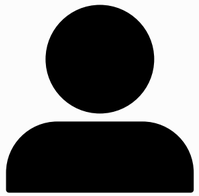
IPCEI-CIS Reference Arquitecture (ICRA)



GSMA: East-Westbound Interface APIs

Functional Requirements

FR1



Developer



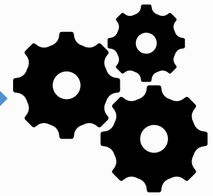


Provider Acess Point (PAP)

FR6



PAP

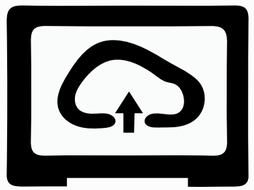


Federation Manager



Request servido localmente

Request enviado a outra operadora



Running Application

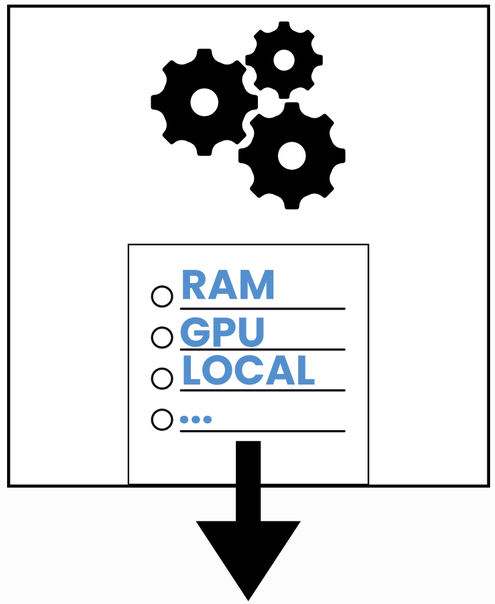


Hand-over

Functional Requirements

FR8

Federation Manager

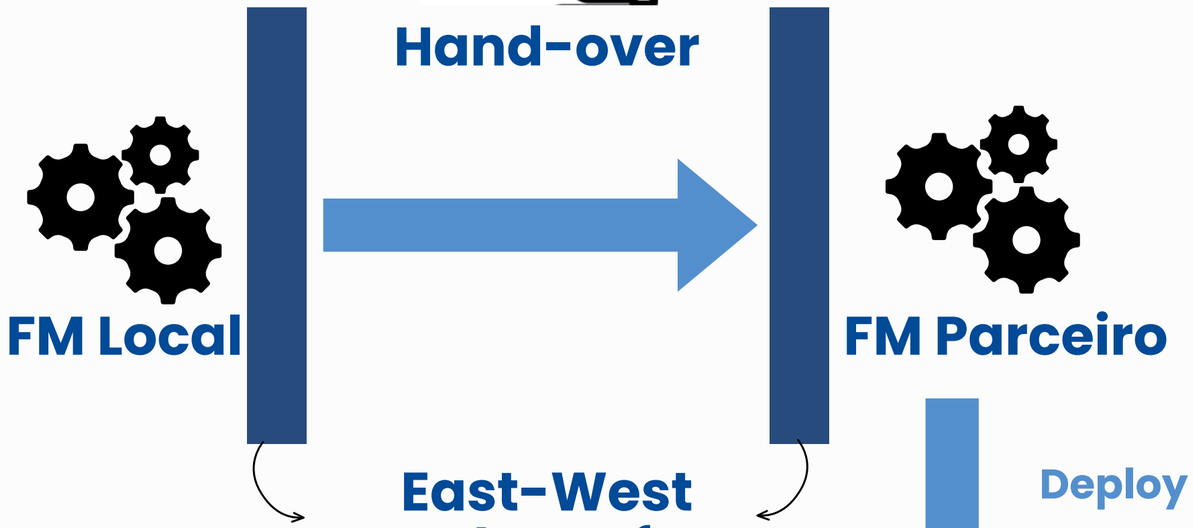


Federation Catalogue

Sabe Operadoras Parceiras

FR5

Hand-over



Running Application

Non-Functional Requirements

NFR 4, 5, 6
Performance



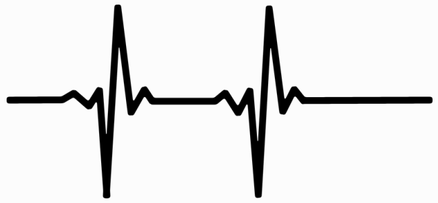
Decisão Deploy Local
<1 s



Hand-over completo (applicação em estado running)
<60 s



Status reports, fora tempo de deployment
<500 ms



Non-Functional Requirements

NFR 8
Resilience



Deve recuperar perda de conexão do estado da aplicação

<30 s



NFR 9, 10
Privacy



Aplicações devem correr isoladas



Operadoras partilham o mínimo de informação possível entre si



Federation Catalogue



Request Data

Personas



Sofia, 29

Developer

Deploys App using PAP



Lau, 26

Home Operator Provider

Handover



João, 50

Partner Operator Provider

Runs Workload

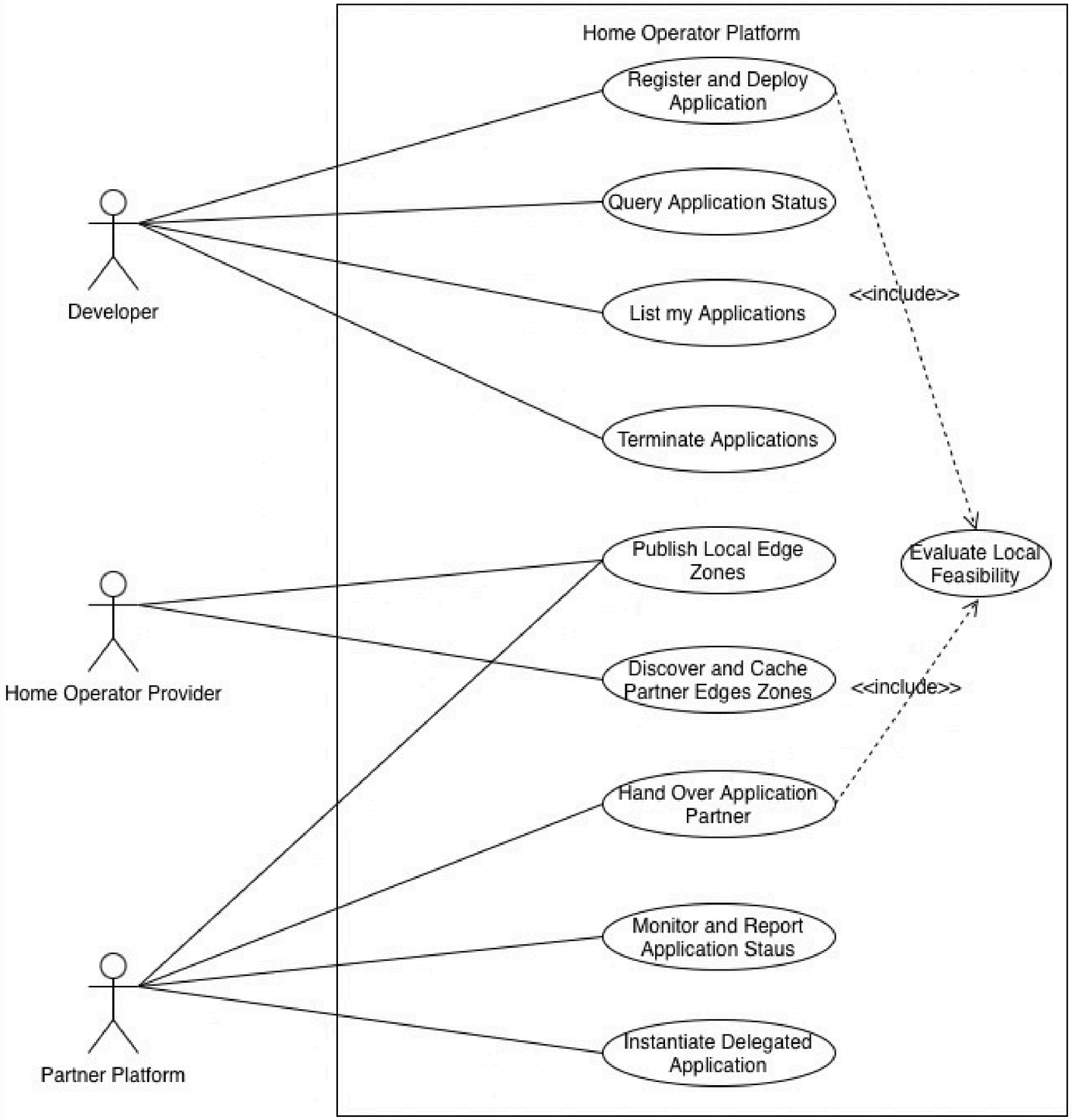


Valério, 34

End User

USE CASES

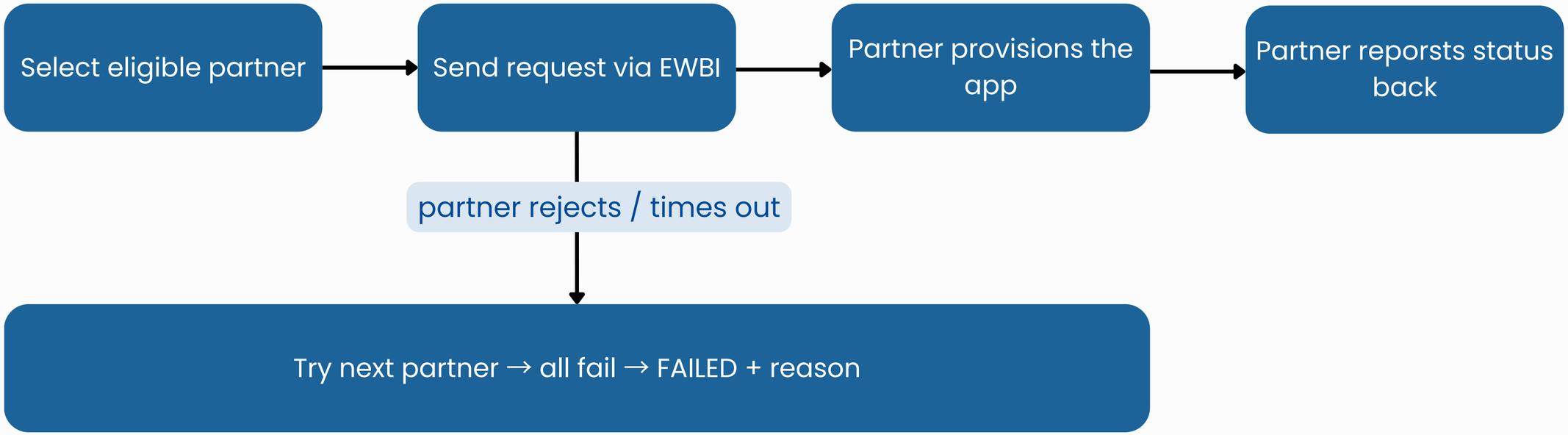
3 actors
9 use cases
5 packages



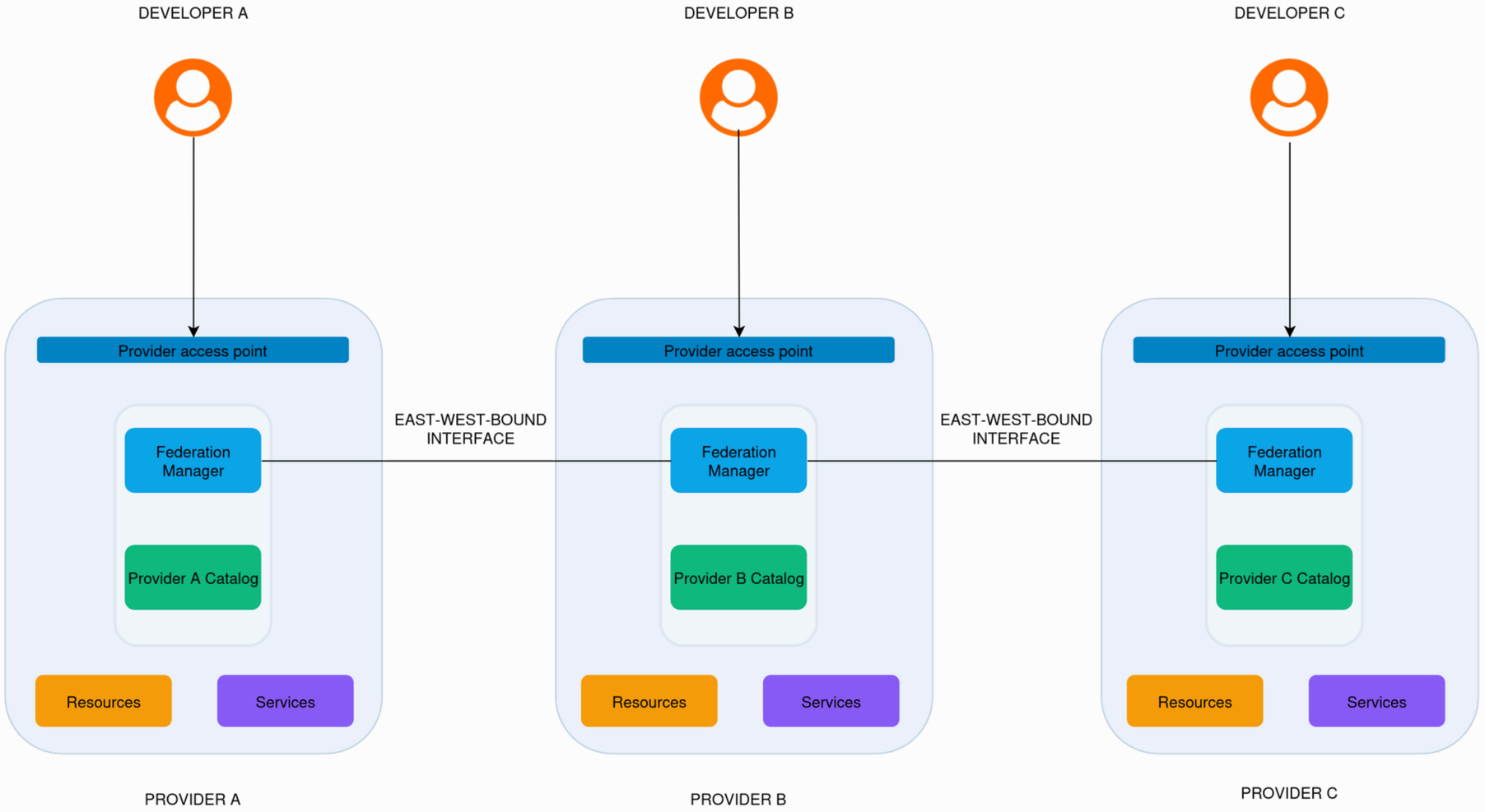
USE CASES

UC7: Hand Over Application to Partner

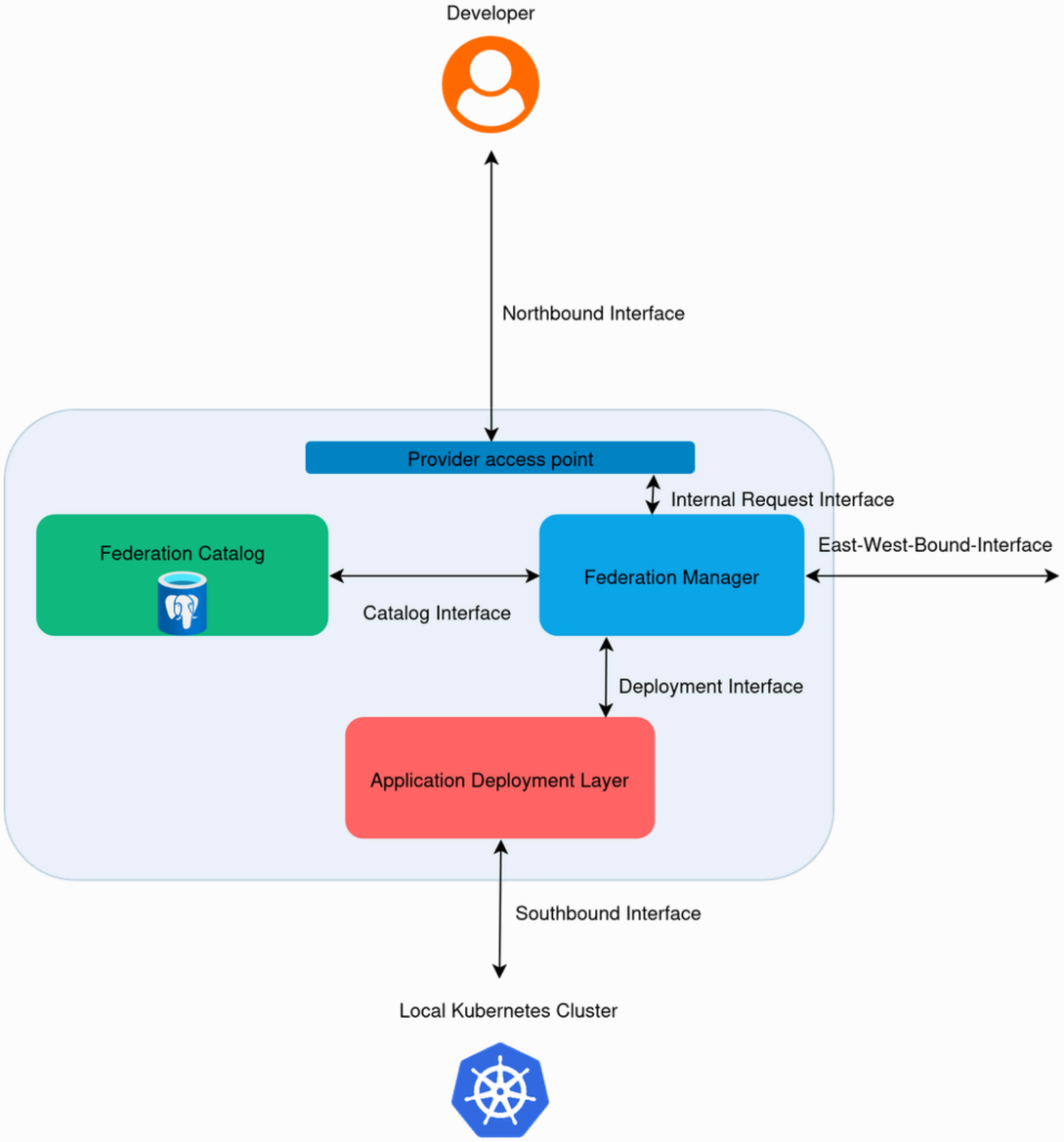
Actors: Federation Manager ↔ Partner Platform



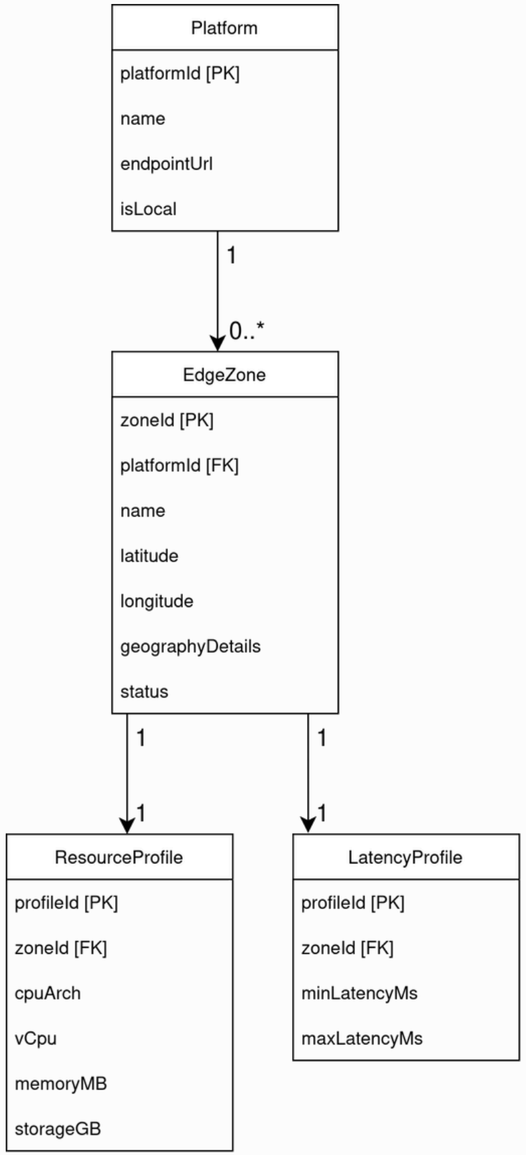
High-Level Architecture



Component Architecture



Catalog Domain Model



MOCKUP

Developer

Deploy Application

Confirm deployment details and review expected latency & cost before

Image Tag

e.g., app/frontend:v1.4.2

Use semantic tags: branch, semantic version (v1.4.2) or canary. Avoid ephemeral hashes for stable releases.

Target Region

us-east-1 (Virginia)

Choose the closest region to your users for lowest edge latency. Some regions may incur additional cost.

SLA Target (ms)

75 Acceptable range 10 - 500 ms

Validation: within allowed range

If SLA not met, automatic rollback may occur.

Deployment Notes

This modal will trigger a rolling edge deploy across selected regions. Estimated time: 3–7 minutes depending on traffic and image size.

Deployment Summary Preview

75 ms

Expected median latency

Estimated Cost

$1,240 /mo $0.058/hr

Throughput	Redundancy
~12k RPS	Multi-zone



Preview of edge footprint and active regions for this deployment.

Latency (last 60m)



Avg: 78 ms P95: 142 ms

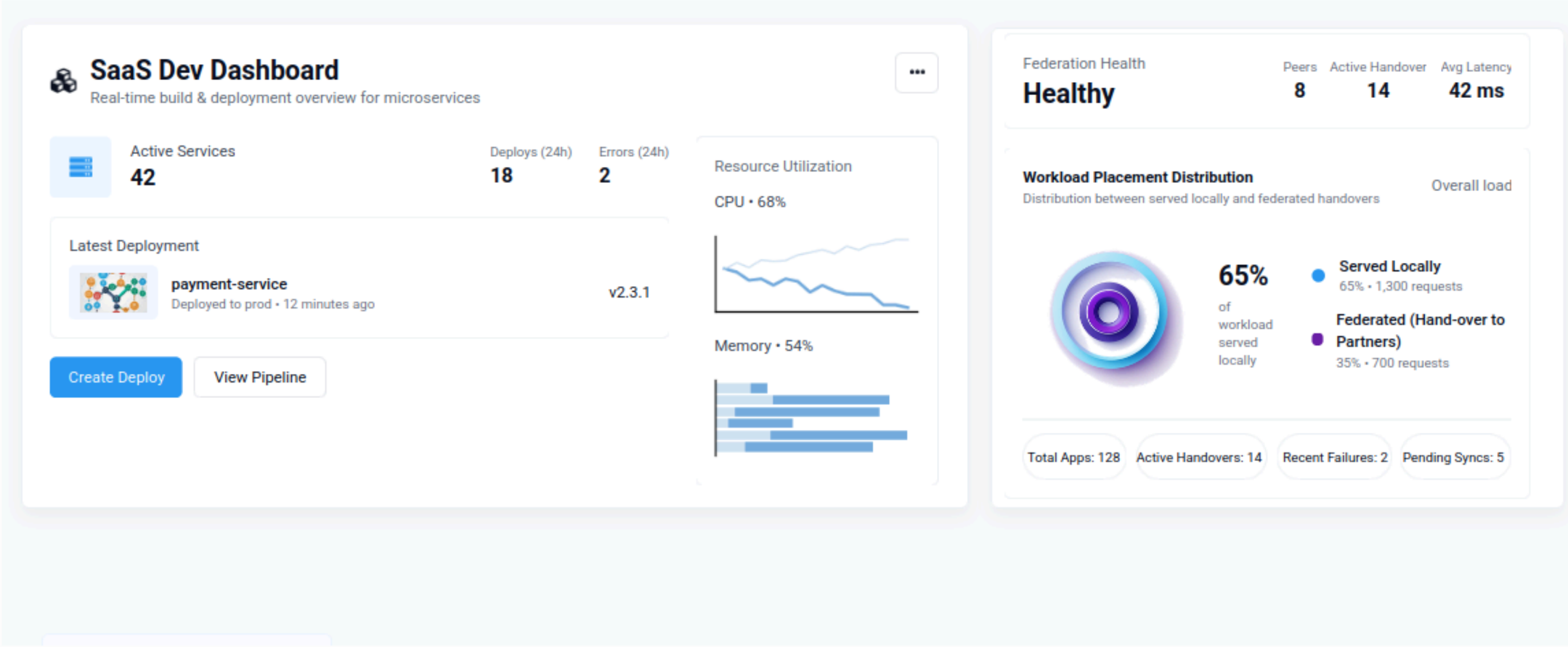
running pending failed

View advanced options

Cancel Confirm Deploy

MOCKUP

Home Provider Operator



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Thank You